# When Does Egocentric Human Data Help Robot Policies? A Spectral Theory of Cross-Embodiment Data Mixing

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## Abstract

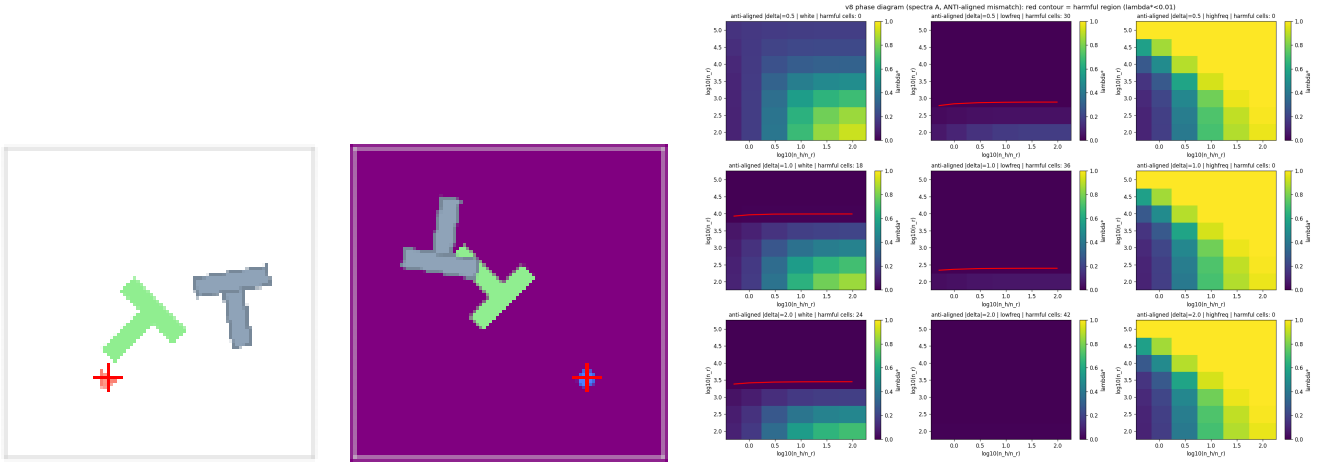
*Robot demonstration data is expensive to collect; egocentric human data is cheap and abundant. Mixing the two is increasingly common in practice, yet the design rules are empirical folklore: reported substitution ratios of 4–5 ego-hours per teleoperation-hour, occasional negative transfer, and no principled answer to* when *and* how much *egocentric data helps a robot policy. We give a first-principles account. We model the two domains as Gaussian covariates with power-law feature spectra, the embodiment gap as a task-vector mismatch decomposed in the shared spectral basis, and the policy fit as mixture-weighted ridge regression evaluated on robot-domain risk. Our main theorem characterizes the mixture risk in closed per-coordinate form (Monte-Carlo verified to 1–2% relative error) and reveals a three-term structure of the marginal benefit of human data: an alignment term, a variance-reduction term, and a spectral-coverage term. From it we derive an exact phase-transition criterion: mixing is beneficial at all robot sample sizes iff a functional $H$ of the mismatch spectrum and the spectra is non-positive; otherwise a critical robot sample size $n_r^{\mathrm{crit}} = K/H$ appears beyond which human data should be discarded — recovering the classical source-discard transition as regularization vanishes. An exchange-rate corollary defines $\kappa_{\mathrm{ex}}$, the number of robot demonstrations equivalent to one egocentric demonstration, shows it is increasing in the inter-domain spectral gap and capped at 1, and lands in the same range as empirically reported substitution ratios. We validate the structural predictions in a controlled dual-domain Push-T testbed with diffusion policies: the first batch confirms the predicted non-monotone benefit with an interior optimum (up to $+32\%$ closed-loop coverage) and harm boundary, shows the benefit grows with training budget and that alignment strength is itself non-monotone, and surfaces an open-loop/closed-loop divergence as a target for the next theory iteration.*

## 1. Introduction

Robot learning is data-hungry, and its current fuel — teleoperated robot demonstrations — is expensive, slow to collect, and hard to scale. Egocentric human data offers a tempting substitute: wearable-camera recordings of humans performing manipulation tasks are cheap, abundant, and semantically aligned with the tasks we want robots to do. A growing line of systems therefore mixes egocentric human data into robot policy training, using hand-mounted grippers [14], hand-held interfaces [6], portable motion-capture rigs [23], or simulation surrogates of the human embodiment [21], and reports substantial gains.

Yet the practice is ahead of the principle. Reported pipelines rely on rules of thumb — fixed mixing ratios, substitution rates on the order of 4–5 ego-hours per teleoperation-hour [7], alignment losses tuned by trial and error — and the same recipe that helps in one setting degrades the policy in another through negative transfer and gradient conflict. Classical domain-adaptation theory [2, 25] explains *why* source data can help or hurt, but its bounds are existential rather than operational: they do not tell a practitioner what the optimal mixing ratio is, when mixing is strictly harmful, or what one egocentric demonstration is worth in units of robot demonstrations.

This paper takes a step toward such an operational theory, and validates it in a controlled testbed. We abstract cross-embodiment data mixing into a *dual-domain model*: observations in each domain are Gaussian covariates with power-law feature spectra $\lambda_j^e \asymp j^{-\alpha_e}$ — motivated by the heavy-tailed spectra ubiquitously observed in learned visual representations [18] — the two domains share a spectral basis, and the embodiment gap is decomposed into a task mismatch vector $\delta$, an alignment coefficient $c = \delta^\top \beta^*$, and per-domain noise. Policies are fit by mixture-weighted ridge regression and evaluated by excess risk on the robot domain. The model is deliberately simple — linear labels, ridge estimation — because its role is to supply *falsifiable structural predictions* (signs, phase boundaries, monotone directions), which we then test on a diffusion policy in a



(a) Robot domain (target) (b) Surrogate ego domain (source) (c) Theory: where mixing hurts

Figure 1. **Cross-embodiment data mixing, modeled.** (a,b) Our controlled dual-domain Push-T testbed pairs a robot domain with a surrogate egocentric-human domain that differs in appearance, actuator, friction, and task geometry (mirrored T-block configurations). (c) The theory predicts *where mixing hurts*: optimal mixing ratio $\lambda^*$ over the $(n_r,\, n_h/n_r)$ plane under anti-aligned mismatch; low-frequency-concentrated mismatch collapses $\lambda^* \to 0$ (dark region), while the same mismatch energy placed on high-frequency coordinates is harmless everywhere (Sec. 4).

controlled dual-domain Push-T environment rather than to numerically match a deep learner.

Our findings are fourfold. First, the mixture risk admits a closed per-coordinate characterization (Theorem 1), exact up to a relative error of $O(n_r^{-3/2})$ and verified numerically to 1–2%; the analysis exposes a *label-feedthrough* effect in which out-of-coordinate signal energy acts as effective noise, $\tilde{\sigma}_e^2 = \sigma_e^2 + F_e$, which is essential for quantitative accuracy. Second, the marginal benefit of human data decomposes into three interpretable terms — alignment, variance reduction, and spectral coverage — whose competition yields an exact phase-transition criterion (Theorem 3): mixing is marginally beneficial at *all* robot sample sizes iff $H \le 0$ for an explicit functional $H$; otherwise a critical threshold $n_r^{\mathrm{crit}} = K/H$ appears, and the classical discard-the-source transition [2] is recovered exactly as regularization vanishes. Strikingly, whether mixing hurts is governed less by the *magnitude* of the mismatch than by the *location of its spectrum* relative to the risk-weighting spectrum: mismatch concentrated on low-frequency, high-risk-weight coordinates is harmful almost everywhere, while the same energy on high-frequency coordinates is harmless (Fig. 1). Third, an exchange-rate corollary (Corollary 4) defines $\kappa_{\mathrm{ex}}$, the number of robot demonstrations equivalent to one egocentric demonstration, and shows it increases with the spectral gap $\alpha_r - \alpha_h$, decreases with mismatch and with $n_r$, and is capped at 1 — beyond which human data covers directions that no amount of additional robot data reaches. The predicted range $\kappa_{\mathrm{ex}} \approx 0.2$–$0.6$ in the small-gap regime quantitatively matches empirically reported 4–5:1 substitution ratios [7]. Fourth, we build a controlled dual-domain Push-T testbed (Sec. 5) in which every gap component (appearance, actuator, friction, task geometry, coverage) is independently manipulable, and run a first batch of mixture experiments with a diffusion policy (Sec. 6). The experiments confirm the theory's three headline structural predictions: naive mixing yields a *non-monotone* benefit curve with an interior optimum ($\lambda^* \approx 0.5$, $+28\%$ closed-loop coverage) and a harm boundary at $\lambda \to 1$; the benefit *grows* with training budget ($+32\%$ at 150 epochs, relieving overfitting exactly as the variance term predicts); and alignment strength is itself non-monotone — light MMD alignment helps while strong alignment hurts at intermediate $\lambda$, consistent with the coverage term's logic. They also surface a sharp open-loop/closed-loop divergence: open-loop robot-domain prediction error keeps improving well past the closed-loop optimum, evidence that prediction and execution must be bridged explicitly. The optimum's quantitative position (predicted $\lambda^* \approx 0.12$ vs. measured $\approx 0.5$) is off, and we state the gap and its sources plainly.

**Contributions.** (i) A dual-domain spectral model of cross-embodiment data mixing with an explicit mismatch decomposition (Sec. 3). (ii) A closed-form risk characterization of the mixture estimator with provable error control, verified to 1–2% (Sec. 4, Theorem 1). (iii) An exact phase-transition criterion for harmful mixing (Theorem 3) and an exchange-rate corollary (Corollary 4) that explains empirical substitution ratios. (iv) A controlled dual-domain Push-T testbed and a first experimental batch testing the theory's

structural predictions with diffusion policies (Secs. 5–6). Proofs, numerical verification protocols, and experimental details are in the supplementary material.

## 2. Related work

**Egocentric human data for robot learning.** Egocentric video has long served as a representation prior: Ego4D [10] supplies large-scale first-person video, and R3M/VIP-style pretraining [17, 19] distills it into visual features for downstream control. A more recent line injects egocentric data *directly into policy training*: EgoMimic [14] uses hand-mounted grippers whose kinematics match the robot, UMI [6] uses hand-held grippers with visual-inertial pose recovery, DexCap [23] uses portable mocap, and EgoBridge [21] studies mixing in simulation with explicit domain gaps. Cross-embodiment robot corpora [20] face a cousin of the same problem across robot morphologies. These works establish that mixing *can* help and document substitution ratios [7] and failure modes, but stop short of a principled account of when and why — the gap this paper addresses.

**Domain adaptation and mixture theory.** Classical domain-adaptation theory bounds target error by source error plus a discrepancy and a shared-optimal-hypothesis term [2]; later work asks when quantity compensates for quality [3], characterizes the value of target data via transfer exponents [12], and shows that enforced representation invariance can provably *hurt* when label marginals differ [25]. On the practice side, data-mixture scaling laws [4] and analyses of sim-real co-training [15] give empirical-fit models of multi-source learning. Closest to us is the ridge-regression scaling analysis of [4], whose minimax characterization assumes matched task vectors across domains; our model adds an explicit mismatch spectrum, which is precisely the object that governs harmful mixing (Theorem 3). Relative to all of the above, our contribution is an *operational* theory: closed-form mixture risk, an explicit critical-threshold formula, and an exchange rate, all stated in quantities estimable from data.

**Technical tools.** Our analysis builds on the ridge-regression literature under power-law spectra [8, 13], with two differences that matter: the robot-domain risk is $\Sigma_r$-weighted (only target-geometry errors count), and the sample-level coupling between coordinates induces a *label-feedthrough* term that acts as effective noise (Sec. 3). On the alignment side we use kernel MMD [11, 16] as the alignment-strength knob in experiments; gradient-conflict diagnostics follow the multi-task learning literature [24]; and the impossibility caveats of invariant-representation approaches [1, 25] inform our negative results.

## 3. A dual-domain model of cross-embodiment mixing

We abstract the problem to its statistical skeleton. Two domains provide training data: the *robot domain* $r$ (target; teleoperated demonstrations) and the *human domain* $h$ (source; egocentric recordings). We write $e \in \{r, h\}$ for the domain index.

**Assumption 1** (Data model)**.** Coordinates $j = 1, \dots, p$ index a shared spectral basis. Observations are $x_e = \Sigma_e^{1/2} u_e$ with $u_e$ standard Gaussian and $\Sigma_e = \mathrm{diag}(\lambda_j^e)$ diagonal with power-law spectra

$$\lambda_j^e \asymp j^{-\alpha_e}, \qquad \lambda_1^e = 1, \tag{1}$$

and throughout we abbreviate $a_j = \lambda_j^r$, $b_j = \lambda_j^h$.

**Assumption 2** (Labels and mismatch)**.** $y_r = \langle \beta^*, x_r \rangle + \varepsilon_r$ in the robot domain; $y_h = \langle \beta^* + \delta, x_h \rangle + \varepsilon_h$ in the human domain. The *task mismatch* $\delta$ absorbs retargeting error, kinematic differences, and residual appearance-induced shifts; its *alignment* with the task is $c = \delta^\top \beta^*$. Noise variances are $\sigma_e^2$, and $\beta^*$ obeys an ellipsoid constraint $\sum_j j^{2s} \beta_j^{*2} \le 1$ with $s > 0$.

**Assumption 3** (Mixture estimator)**.** Given $n_r$ robot and $n_h$ human samples and a mixing ratio $\lambda \in [0, 1]$, per-sample weights are $w_r = (1 - \lambda)/n_r$, $w_h = \lambda/n_h$. We analyze the *diagonalized ridge* estimator

$$\hat{\beta}_j = \frac{w_r T_{rj} + w_h T_{hj}}{w_r S_{rj} + w_h S_{hj} + \mu}, \qquad \mu > 0, \tag{2}$$

with $S_{ej} = \|X_{e,:j}\|^2$, $T_{ej} = X_{e,:j}^\top y_e$; $\mu$ is the ridge penalty and stands in for explicit or implicit regularization (weight decay, early stopping).

The evaluation criterion is the *robot-domain excess risk*, weighted by the robot geometry:

$$\mathcal{R}_T(\lambda) = \mathbb{E} \left\| \hat{\beta}(\lambda) - \beta^* \right\|_{\Sigma_r}^2. \tag{3}$$

Only errors in directions the robot actually visits are penalized — this weighting turns out to be essential for the phase structure in Sec. 4.

**Label feedthrough.** A subtle point drives the quantitative accuracy of the theory. Although the population covariances are diagonal, the *sample* Gram matrices are not: the label projection $T_{ej}$ picks up contributions from all other coordinates, $T_{ej} = S_{ejj} \beta_{e,j} + \sum_{k \neq j} S_{ejk} \beta_{e,k} + X_{e,:j}^\top \varepsilon_e$. The middle term is mean-zero with variance $n_r \lambda_j^e F_e$ (and similarly for $h$), where $F_e = \sum_k \lambda_k^e \beta_{e,k}^2$ is the domain's

total signal energy. Out-of-coordinate signal is therefore indistinguishable from noise, giving rise to the *effective noise*

$$\tilde{\sigma}_e^2 = \sigma_e^2 + F_e. \tag{4}$$

A heavier-tailed human spectrum (small $\alpha_h$) inflates $F_h$: broad coverage carries a hidden per-coordinate SNR cost, which competes with its coverage benefit — this tension is the microscopic origin of the phase diagrams in Sec. 4.

**What the model keeps and discards.** The model retains the three gap components that mixture practice manipulates — spectra (coverage), mismatch $\delta$ (embodiment/retargeting), and noise — and discards the rest (nonlinear observation maps, temporal structure, closed-loop compounding). Sec. 6 therefore tests only *structural* predictions: signs, monotone directions, and phase-region membership, not pointwise numerical values. The diagonalization in Assumption 3 is a controlled approximation of the full ridge solution; we quantify its gap (8–28% on the weighted risk) in the supplementary and prove the full-ridge deterministic equivalent there in the fixed-dimension regime.

## 4. Theory

All proofs and the full numerical-verification protocol are deferred to the supplementary; here we state results and give intuition.

### 4.1. Closed-form risk of the mixture estimator

**Theorem 1** (Risk characterization, T1-a)**.** *Under Assumptions 1–3, with $n_r, n_h \to \infty$ at fixed ratio $\rho = n_h/n_r$ and $p$ fixed or growing polynomially, with high probability,*

$$\left|\mathcal{R}_T(\lambda) - \bar{\mathcal{R}}_T(\lambda)\right| \;\le\; C\big(\kappa_0^{-3} + n_r^{-3/2}\big)\,\bar{\mathcal{R}}_T(\lambda), \tag{5}$$

*where the deterministic equivalent is*

$$\bar{\mathcal{R}}_T(\lambda) = \sum_{j>m} a_j \beta_j^{*2} + \sum_{j\le m} a_j\,\mathrm{MSE}_j^{\mathrm{det}}(\lambda), \tag{6}$$

*with per-coordinate closed form*

$$\mathrm{MSE}_j^{\mathrm{det}} = \underbrace{\frac{(1-\lambda)^2\tilde{\sigma}_r^2 a_j/n_r + \lambda^2\tilde{\sigma}_h^2 b_j/n_h}{D_j^2}}_{\textit{effective variance}} + \underbrace{\Big(\frac{\lambda\, b_j\delta_j - \mu\beta_j^*}{D_j}\Big)^2}_{\textit{bias square}}, \tag{7}$$

*where the bias square expands into mismatch energy, shrinkage energy, and an* alignment cross term *$-2\lambda\mu b_j\delta_j\beta_j^*/D_j^2$ whose sign tracks the alignment $c$; $D_j = (1-\lambda)a_j + \lambda b_j + \mu$, $\tilde{\sigma}_e^2 = \sigma_e^2 + F_e$ the feedthrough-corrected effective noise of Eq. (4), and $m$ a self-consistent truncation $m = \max\{j : D_j \ge \kappa_0\eta_j\}$ with $\eta_j$ the fluctuation scale of the empirical denominator. Second-order fluctuation corrections to Eq. (7) are derived in the supplementary and included in all numerical evaluations.*

**Proof sketch.** The chain is elementary once the feedthrough is identified: (i) a per-coordinate representation of the estimator including the feedthrough term (Lemma L1); (ii) exact conditional MSE given the design (Lemma L2); (iii) a deterministic equivalent via $\chi^2$ concentration of $S_{ejj}$ and a second-order delta method, in which the only third-order mixed moment closes in the exact form $8\lambda^3 b_j^3\delta_j^2/n_h^2$ (Lemma L3); (iv) assembly with a truncation at the information-poor coordinates, where a boundary-band argument keeps the aggregate relative error at $O(\kappa_0^{-3})$ (Lemma L4). Every step is scalar random-matrix-free analysis; no new asymptotic-equivalence lemma is needed in the diagonal setting.

**Numerical verification.** Across spectral families and mixing ratios, Eq. (7) matches Monte-Carlo estimates with binned median relative error 0.6–1.6% and aggregate error $\le$ 2% (200 replicates, ∼200k coordinate-replicates; the supplementary material). Against the *full* (non-diagonalized) ridge estimator, Eq. (6) is a variance upper bound; the gap on the weighted risk is 8–28% depending on the spectral family, and the full-ridge extension (T1-b) is proved in the supplementary for the fixed-dimension regime.

### 4.2. When does marginal mixing help? Three forces

**Corollary 2** (Marginal-benefit decomposition, C1)**.** *The right derivative $\bar{\mathcal{R}}_T'(0^+) = \sum_j a_j\,\partial_\lambda \mathrm{MSE}_j^{\mathrm{det}}(0)$ decomposes per coordinate as (sign flipped so that positive means beneficial)*

$$\mathrm{benefit}_j \;\propto\; \underbrace{\mu\beta_j^* b_j\delta_j}_{\textit{A: alignment}} + \underbrace{V_j D_{0j}}_{\textit{B: variance}} + \underbrace{(V_j + \mu^2\beta_j^{*2})(b_j - a_j)}_{\textit{C: coverage}}, \tag{8}$$

*with $V_j = \tilde{\sigma}_r^2 a_j/n_r$ and $D_{0j} = a_j + \mu$.*

The three terms have distinct physical carriers. A (alignment) exists only through $\mu$: regularization is what couples the mismatch to the task direction — with sign set by $c = \delta^\top\beta^*$. B (variance) is always positive and vanishes as $n_r$ grows: diminishing returns. C (coverage) is positive exactly where the human spectrum is heavier than the robot spectrum, $b_j > a_j$; it is the spectral fingerprint of "the human data sees directions the robot data cannot". In the unregularized limit only B survives:

$$\bar{\mathcal{R}}_T'(0^+)\big|_{\mu=0} = -\frac{2\tilde{\sigma}_r^2}{n_r}\sum_j j^{\alpha_r - \alpha_h} < 0 \quad \text{always}, \tag{9}$$

so under-regularized mixture is always beneficial, with magnitude growing in the spectral gap — the classical picture.

### 4.3. The phase transition: an exact criterion

**Theorem 3** (Harmful-mixing criterion, C2)**.** *Define the asymptotic marginal-harm functional (the $n_r \to \infty$, $n_h/n_r$ fixed part of $\bar{\mathcal{R}}'_T(0^+)$ after term B vanishes), for anti-aligned mismatch:*

$$H = \underbrace{2\mu \sum_j \frac{a_j \beta_j^* b_j |\delta_j|}{D_{0j}^2}}_{A_\infty:\ \text{mismatch harm}} - \underbrace{2\mu^2 \sum_j \frac{a_j \beta_j^{*2}(b_j - a_j)}{D_{0j}^3}}_{C_\infty:\ \text{coverage benefit}}. \tag{10}$$

*Then (i) $H \le 0$ implies marginal mixing is beneficial at* all *robot sample sizes; (ii) $H > 0$ implies a critical robot sample size*

$$n_r^{\text{crit}} = K/H, \qquad K = 2\tilde{\sigma}_r^2 (S_1 + S_2), \tag{11}$$

*with $S_1 = \sum_j a_j^2/D_{0j}^2$, $S_2 = \sum_j a_j^2(b_j - a_j)/D_{0j}^3$, beyond which marginal mixing hurts.*

**Exactness.** The criterion is not an asymptotic slogan. Numerically (the supplementary material): the predicted mismatch threshold $\|\delta\|^* = 0.674$ (the root of $H$) coincides with the Monte-Carlo phase boundary to within the search resolution, and $n_r^{\text{crit}} \cdot H = 25.866$ is constant across three decades of mismatch magnitude — the inverse proportionality in Eq. (11) is exact.

**Two consequences.** First, *mismatch location beats mismatch magnitude*: because $A_\infty$ weights $|\delta_j|$ by the risk weight $a_j \beta_j^* b_j / D_{0j}^2$, mismatch concentrated on low-frequency, high-risk-weight coordinates is harmful across nearly the whole $(n_r, n_h/n_r)$ plane (42/42 cells at $\|\delta\| = 2$), while the same energy on high-frequency coordinates is harmless everywhere (0/42) — see Fig. 1c and Table 1. Second, *$\mu$ is the carrier of the classical transition*: as $\mu \to 0$, $C_\infty \to 0$ and $H = A_\infty > 0$ whenever $c < 0$, recovering the discard-the-source transition of classical domain adaptation; at $\mu > 0$ the large-$n_r$ limit of the mixing benefit is $-C_\infty < 0$ — broad human coverage keeps paying even with abundant robot data.

### 4.4. The exchange rate of egocentric data

**Corollary 4** (Exchange rate, T2)**.** *Define the ego↔teleop exchange rate $\kappa_{\text{ex}}$ by*

$$\bar{\mathcal{R}}_T(\lambda^*;\, n_r, n_h) = \bar{\mathcal{R}}_T\big(0;\, n_r + \kappa_{\text{ex}}\, n_h\big), \tag{12}$$

i.e.*, the amount of extra robot data that reproduces the optimal mixture's risk. Then $\kappa_{\text{ex}}$ is (i) increasing in the spectral gap $\alpha_r - \alpha_h$; (ii) decreasing in the effective mismatch; (iii) decreasing in $n_r$; and (iv) capped: there is a large-gap regime in which $\kappa_{\text{ex}} \ge 1$,* i.e.*, human data supplies coverage that* no *amount of additional robot data reproduces.*

Table 1. Harmful-mixing cells (out of 42 grid points of $(n_r, n_h/n_r)$) under anti-aligned mismatch, by mismatch magnitude and *spectral location*. Location — not magnitude — is the dominant factor; theory: Eq. (10).

| $\|\delta\|$ | white | low-freq | high-freq |
|---|---|---|---|
| 0.5 | 0/42 | 30/42 | 0/42 |
| 1.0 | 18/42 | 36/42 | 0/42 |
| 2.0 | 24/42 | 42/42 | 0/42 |

Numerically (Fig. in suppl.): at zero spectral gap with moderate mismatch and noisier human labels, $\kappa_{\text{ex}} \approx 0.42$–$0.59$ — one ego sample is worth about half a robot sample, the price of the gap; the cap $\kappa_{\text{ex}} = 1$ binds from $\alpha_r - \alpha_h \approx 0.5$–$0.75$ upward. The empirical substitution ratio of 4–5:1 reported for egocentric pipelines [7] corresponds to $\kappa_{\text{ex}} \approx 0.2$–$0.25$ and falls inside the predicted small-gap range 0.19–0.56; we note the units caveat (theory prices samples; empirics price hours). Corollary 4 thus turns a practitioner rule of thumb into a computable function of measurable quantities: two spectra, a mismatch spectrum, and noise levels.

**The $n_h$ axis: an exact dosage law.** The classical small-source threshold has a smooth refinement in this model (supplementary material). At matched spectra the optimal gain is exactly hyperbolic in the human sample size, $g(n_h) = A_1^2/(4A_2(n_h))$ with $A_2(n_h) = A_2^r + B/n_h$; this yields an explicit worth-mixing threshold $n_h^\varepsilon$, a ratio $\lambda^*(n_h)$ growing linearly from zero, and a closed-form exchange rate whose pole is exactly the cap regime of Corollary 4(iv). Unlike the $n_r$ axis — a true sign-change transition (Theorem 3) — the $n_h$ axis carries no sign change: once mixing helps at all, it helps at every sample size, and the hard classical threshold is recovered only as the divergence $n_h^\varepsilon \to \infty$.

## 5. A controlled dual-domain Push-T testbed

Testing the theory requires a testbed in which every component of the embodiment gap is independently manipulable — something real egocentric datasets do not allow. We therefore build a *dual-domain* variant of Push-T, following the simulation-side design of EgoBridge [21] with three deliberate deviations (stated below), on top of the diffusion-policy codebase of [5].

**Domain construction.** Both domains share the Push-T task (push a T-shaped block to a goal pose with a point-mass pusher, image observations) and differ along four

Table 2. Dual-domain Push-T construction. The robot domain is the training+evaluation target; the human domain is the mixture source (a *surrogate* for egocentric data, in the spirit of [21]).

| | robot (target) | human (source) |
|---|---|---|
| pusher | triangle | circle |
| background | white | purple |
| friction | $\times 0.7$ | $\times 1.0$ |
| T-block geometry | canonical | mirrored |
| demonstrations | 122 | 128 |
| paired subset (same init. config) | 102 | |

axes, summarized in Table 2 and Fig. 1: the pusher embodiment (triangle vs. circle), the appearance (white vs. purple background), the contact dynamics (friction scaled $\times 0.7$ vs. unchanged), and the task geometry (the T-block's initial and goal poses are mirrored about the vertical axis in the human domain, so that optimal pushing actions are *not* directly transferable — a controlled kinematic gap). The four axes map onto the model's quantities: appearance/friction differences reshape the two spectra $(a_j, b_j)$; the mirrored geometry induces the task mismatch $\delta$; the pusher change alters both.

**Data.** We replay the 206 expert demonstrations of the public Push-T dataset [5, 9] through both domain simulators and keep the successful rollouts: 122 robot-domain and 128 human-domain demonstrations (14.5k/15.5k steps), of which 102 form *paired* episodes sharing the same initial configuration — the pairs make the mismatch spectrum $\delta_j$ directly estimable from action residuals. A mirror-convention subtlety is worth recording: the physically correct parity transform maps the block angle as $\theta \mapsto -\theta$ (reflection about the handle axis), not $\theta \mapsto \pi - \theta$; the wrong convention silently destroys replay success (0/206 vs. 206/206). Deviations from [21]: our demonstrations are replayed from public expert data rather than generated by a scripted planner; friction differs across domains in our construction; and we add the paired subset for mismatch estimation.

**Policy and training.** The policy is the CNN-based diffusion policy of [5] (2 observation steps, 8 action steps), with a *mixture dataset* that samples human-domain sequences with probability $\lambda$ and robot-domain sequences with $1-\lambda$; validation is always on the robot domain. Alignment strength is controlled by an optional Gaussian-kernel MMD penalty [11] on the shared visual encoder's features, driven by domain labels, with weight $\alpha_{\text{mmd}}$; $\alpha_{\text{mmd}} = 0$ is naive mixing.

Table 3. Closed-loop robot-domain coverage (single seed, coarse batch). Left block: 50 epochs; right block: 150 epochs. The naive curve is *non-monotone with an interior optimum* $\lambda^* \approx 0.5$; the long arm shows the benefit survives — and grows — with training.

| $\lambda$ | naive | $\text{mmd}_{0.1}$ | $\text{mmd}_{1.0}$ | $\text{naive}_{150}$ |
|---|---|---|---|---|
| 0.0 | 0.524 | 0.487 | 0.629 | 0.578 |
| 0.1 | 0.543 | — | — | |
| 0.2 | 0.593 | — | — | |
| 0.35 | 0.629 | 0.644 | 0.475 | 0.763 |
| 0.5 | **0.670** | — | — | |
| 0.7 | 0.587 | 0.624 | 0.464 | 0.650 |
| 0.9 | 0.492 | — | — | |
| 1.0 | 0.118 | 0.141 | 0.127 | — |

**Estimating the model's parameters on the testbed.** The theory's quantities are measured from the dual-domain data: spectra $\hat{\alpha}_r, \hat{\alpha}_h$ by PCA on state-action trajectories (mid-range log-log fit); the task-energy spectrum $\hat{\beta}^{*2}$ by ridge-regressing actions on states in the robot domain; the mismatch spectrum $\hat{\delta}$ from paired action residuals, yielding $\Delta_\varphi = \|\hat{\delta}\|^2$ and $c = \hat{\delta}^\top \hat{\beta}^*$; effective noise $\tilde{\sigma}_e^2$ from regression residuals plus feedthrough energy. On our testbed: $\hat{\alpha}_r = 1.89$, $\hat{\alpha}_h = 1.92$ (near-zero spectral gap), and $c < 0$ — the mirrored construction is indeed *anti-aligned*, so the testbed sits in the regime where the theory predicts only a small marginal benefit, $\lambda^* \approx 0.12$.

## 6. First experiments

**Protocol.** We run a deliberately coarse first batch on the testbed of Sec. 5: *naive* mixing over $\lambda \in \{0, 0.1, 0.2, 0.35, 0.5, 0.7, 0.9, 1.0\}$ (50 epochs), *MMD-aligned* mixing with $\alpha_{\text{mmd}} \in \{0.1, 1.0\}$ over $\lambda \in \{0, 0.35, 0.7, 1.0\}$ (50 epochs), and a *long-training* arm at 150 epochs over $\lambda \in \{0, 0.35, 0.7\}$; $n_r = 80$ robot + $n_h = 100$ human demonstrations, one seed per cell (coarse protocol; a multi-seed batch follows). Per run we record closed-loop robot-domain coverage (mean final T-coverage, 20 evaluation episodes, seeds 101–120; binary success at the 0.855 threshold where attained), open-loop robot-domain validation loss, and three checkpoint probes: latent inter-domain $W_2$, linear domain-classification AUC, and the robot/human gradient cosine with induced $\kappa(\hat{G})$. The theory overlay plugs the testbed parameter estimates ($\hat{\alpha}_r = 1.89$, $\hat{\alpha}_h = 1.92$, $c < 0$) into Eq. (6); we test structural predictions only.

**Findings. (F1) Non-monotone benefit with an interior optimum — P2 confirmed at coarse level.** Naive mixing improves closed-loop coverage over the pure-robot baseline across $0.1 \le \lambda \le 0.9$, peaking at $\lambda^* \approx 0.5$ (0.670 vs. 0.524, $+28\%$ relative), and collapses at $\lambda = 1$ (0.118)

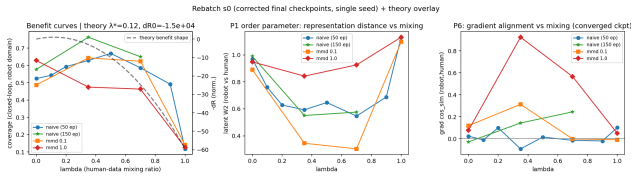



Figure 2. **First batch vs. theory.** Left: closed-loop coverage vs. mixing ratio for the four arms (naive 50-epoch, naive 150-epoch, two MMD weights), with the theoretical benefit shape (dashed; normalized shape comparison only). Center: latent-space inter-domain $W_2$. Right: robot/human gradient cosine at the converged checkpoint.

where no robot data remains — the harm boundary the theory predicts. **(F2) The benefit is not an under-training artifact — and it relieves overfitting.** At 150 epochs all arms improve ($\lambda$=0: 0.578) and the mixing benefit *grows*: $\lambda$=0.35 reaches 0.763 (+32% over its baseline, binary success rate 0.40 vs. 0.25), $\lambda$=0.7 gives 0.650. The training logs show why: at 150 epochs the pure-robot run overfits (train/val loss 0.005/0.164), while mixed runs sit at 0.008/0.085–0.127 — human data acts as a variance-reduction regularizer, the model's term B. The flat curves one would read off an under-trained budget are an artifact of the training regime, not of the mixing structure. **(F3) Open-loop and closed-loop diverge in opposite directions.** The open-loop robot-domain validation loss improves monotonically with human data all the way to $\lambda = 0.9$ (0.072 $\rightarrow$ 0.036), while the closed-loop coverage peaks at $\lambda^* \approx 0.5$ and already declines at $\lambda = 0.9$ (0.492); at $\lambda = 1$ both collapse (0.227, 0.118). Past the closed-loop optimum, extra human data still helps *predict* robot actions but hurts *executing* them — direct evidence that open-loop risk and closed-loop compounding must be bridged explicitly (in the spirit of [22]), which our current model does not yet cover. **(F4) Alignment strength itself is non-monotone.** At $\lambda = 0.7$, light MMD ($\alpha_{\text{mmd}} = 0.1$) improves over naive (0.624 vs. 0.587), but strong MMD ($\alpha_{\text{mmd}} = 1.0$) *hurts* at intermediate $\lambda$ (0.475/0.464 vs. naive 0.629/0.587). This is qualitatively consistent with the model's coverage term C: over-aggressive alignment collapses precisely the spectral breadth through which human data helps — alignment should be tuned, not maximized. **(F5) Representations track the mixing ratio.** Latent $W_2$ decreases with $\lambda$ in the interior (naive: 0.97 $\rightarrow$ 0.55–0.65) and re-expands at $\lambda = 1$ (1.13); the gradient cosine at the converged checkpoint is near zero throughout, with sign noise at extreme $\lambda$ — the probe records the converged state, so training-time conflict tracking is deferred to the multi-seed batch. **(F6) Probe caveats.** Domain AUC saturates ($\approx$ 1.0) and one checkpoint exhibits an ill-conditioned gradient pair ($\text{mmd}_{1.0}$, $\lambda$=0.35: cos 0.93, $\kappa(\hat{G}) = 2.6$); the next batch uses a capacity-limited probe and training-time gradient logging.

**Consistency with theory, stated honestly.** The theory gets right: the *sign* of the marginal benefit, the *existence of an interior optimum*, and the *harm boundary* at $\lambda \rightarrow 1$. It is quantitatively off on the optimum's position: with the testbed's measured near-zero spectral gap and anti-aligned mismatch, Eq. (6) predicts a small benefit peaked near $\lambda^* \approx 0.12$, while the measured peak sits at $\lambda^* \approx 0.5$ with a substantial amplitude. This gap is expected at two levels: the model is a linear-ridge asymptotic object while the learner is a 50-epoch diffusion policy, and the mismatch/alignment estimates $\hat{\delta}$, $\hat{c}$ come from a linear state-action proxy that need not track the learned policy's effective mismatch. Closing this quantitative gap (calibrated $\mu$, learned-feature spectra, multi-seed significance) is the purpose of the next batch; the structural agreement reported here is the scope we claim in this paper.

## 7. Discussion

**What the theory buys a practitioner.** Three quantities estimable from data — two spectra, a mismatch spectrum, and noise levels — determine (i) whether human data helps at all (sign of Eq. (8)), (ii) whether there is a robot-data scale beyond which it must be discarded (Theorem 3), and (iii) what it is worth in robot-data units (Corollary 4). The mismatch-*location* effect (Table 1) is, to our knowledge, a new and actionable prediction: retargeting errors that live on low-frequency, task-critical coordinates are far more dangerous than equal-magnitude errors on high-frequency coordinates — which suggests concrete filtering strategies for egocentric pipelines.

**Boundaries of the theory.** The theory has a sharp negative edge, stated fully in the supplementary material. If the encoder aligns the two domains in feature space ($W_1$ gap $\varepsilon$) while the action marginals differ (mean gap $|\Delta\mu|$), then *any* deterministic $L$-Lipschitz shared head pays mixture risk $R_\lambda \geq \lambda(1-\lambda)(|\Delta\mu| - L\varepsilon)_+^2$, and the mixture-optimal head pays $R_r \geq \lambda^2(|\Delta\mu| - L\varepsilon)_+^2$ on the robot domain — a floor that *grows* with the mixing ratio, the squared-loss analogue of the bound of [25]. Over-strong alignment losses are likewise predicted to collapse useful within-domain structure, the overlapping/collapse regime of [15]; the first batch observes the predicted signature — strong MMD alignment reduces closed-loop coverage at intermediate $\lambda$ relative to both light alignment and naive mixing (Sec. 6, F4). Finally, with only two embodiment "environments", IRM-style full invariance is unidentifiable [1]; this is why the framework claims only *partial* invariance (appearance-invariant features with a domain-conditioned action head).

**Limitations.** Our theory is stated for a diagonalized ridge estimator over linear labels; the gap to the full ridge solution is 8–28% on the weighted risk (a full-ridge extension is stated as a conjecture in the supplementary), and the gap to diffusion policies is bridged only at the level of structural predictions, which is all we claim. The experiments so far cover one task, one policy class, one surrogate ego domain, and a coarse single-seed protocol; the alignment-strength axis (two MMD weights) and a 150-epoch arm are included, but multi-seed significance, the coverage axis, and action-marginal-shift tests are future work. The exchange rate prices *samples*, while empirical practice prices *hours*; the conversion depends on collection throughput. Finally, the harmful-region criterion assumes the shared-spectral-basis structure of Assumption 1; heavy basis misalignment across embodiments is outside the current model.

**Outlook.** The immediate next steps are set by the theory itself: a multi-seed significance pass over the interior optimum, the alignment-strength response (why does strong alignment hurt, and where is the transition?), the open-loop-to-closed-loop bridge (compounding-error bounds in the spirit of [22]), and a real-robot egocentric pairing of the testbed. More broadly, we hope the exchange-rate formulation turns "how much egocentric data should I mix?" from folklore into a measurable, computable quantity.

**Conclusion.** We gave a closed-form, numerically exact account of cross-embodiment data mixing: a per-coordinate risk theorem, an exact harmful-mixing phase criterion, and an exchange-rate corollary that quantitatively matches empirical substitution ratios. A controlled dual-domain Push-T testbed confirms the theory's structural predictions with diffusion policies and surfaces an open-loop/closed-loop divergence as a target for the next iteration of the theory.

## References

[1] Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization. *arXiv:1907.02893*, 2019. 3, 7

[2] Shai Ben-David, John Blitzer, Koby Crammer, Alex Kulesza, Fernando Pereira, and Jennifer Wortman Vaughan. A theory of learning from different domains. *Machine Learning*, 79(1-2):151–175, 2010. 1, 2, 3

[3] Shai Ben-David, Shai Shalev-Shwartz, and Ruth Urner. Domain adaptation—can quantity compensate for quality? *Annals of Mathematics and Artificial Intelligence*, 2014. 3

[4] Lehan Chen, Diyuan Wu, and Marco Mondelli. When do data mixtures improve scaling laws? insights from high-dimensional regression. *arXiv:2609.38011*, 2026. 3

[5] Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *RSS*, 2023. 5, 6

[6] Cheng Chi, Zhenjia Xu, Chuer Pan, Eric Cousineau, Benjamin Burchfiel, Siyuan Feng, Russ Tedrake, and Shuran Song. Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots. In *RSS*, 2024. 1, 3

[7] Xin Ding, Fu Chen, Bingjia Huang, et al. Zeva-ego: Egocentric mid-training with in-context causal learning for robot manipulation. *arXiv:2609.24411*, 2026. 1, 2, 3, 5

[8] Edgar Dobriban and Stefan Wager. High-dimensional asymptotics of prediction: Ridge regression and classification. *Annals of Statistics*, 46(1):247–279, 2018. 3

[9] Pete Florence, Corey Lynch, Andy Zeng, Oscar A. Ramirez, Ayzaan Wahid, Laura Downs, Adrian Wong, Johnny Lee, Igor Mordatch, and Jonathan Tompson. Implicit behavioral cloning. In *CoRL*, 2021. 6

[10] Kristen Grauman, Andrew Westbury, et al. Ego4d: Around the world in 3,000 hours of egocentric video. In *CVPR*, 2022. 3

[11] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *JMLR*, 13:723–773, 2012. 3, 6

[12] Steve Hanneke and Samory Kpotufe. On the value of target data in transfer learning. In *NeurIPS*, 2019. 3

[13] Trevor Hastie, Andrea Montanari, Saharon Rosset, and Ryan J. Tibshirani. Surprises in high-dimensional ridgeless least squares interpolation. *Annals of Statistics*, 50(2):949–986, 2022. 3

[14] Simar Kareer, Dhruv Patel, et al. Egomimic: Scaling imitation learning via egocentric video. *arXiv:2410.24221*, 2024. 1, 3

[15] Yu Lei, Minghuan Liu, Abhiram Maddukuri, Zhenyu Jiang, and Yuke Zhu. A mechanistic analysis of sim-and-real co-training in generative robot policies. *arXiv:2604.13645*, 2026. 3, 7

[16] Mingsheng Long, Yue Cao, Jianmin Wang, and Michael I. Jordan. Learning transferable features with deep adaptation networks. In *ICML*, 2015. 3

[17] Yecheng Jason Ma, Shagun Sodhani, Dinesh Jayaraman, Osbert Bastani, Vikash Kumar, and Amy Zhang. Vip: Towards universal visual reward and representation via value-implicit pre-training. *arXiv:2210.00103*, 2022. 3

[18] Charles H. Martin and Michael W. Mahoney. Implicit self-regularization in deep neural networks: Evidence from random matrix theory and implications for learning. *JMLR*, 22, 2021. 1

[19] Suraj Nair, Aravind Rajeswaran, Vikash Kumar, Chelsea Finn, and Abhinav Gupta. R3m: A universal visual representation for robot manipulation. In *CoRL*, 2022. 3

[20] Open X-Embodiment Collaboration. Open x-embodiment: Robotic learning datasets and rt-x models. *arXiv:2310.08864*, 2023. 3

[21] Ryan Punamiya, Dhruv Patel, Patcharapong Aphiwetsa, Pranav Kuppili, Lawrence Y. Zhu, Simar Kareer, Judy Hoffman, and Danfei Xu. Egobridge: Domain adaptation for generalizable imitation from egocentric human data. In *NeurIPS*, 2025. 1, 3, 5, 6

[22] Stéphane Ross, Geoffrey J. Gordon, and J. Andrew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *AISTATS*, 2011. 7, 8

[23] Chen Wang, Haochen Shi, et al. Dexcap: Scalable and portable mocap data collection system for dexterous manipulation. *arXiv:2403.07788*, 2024. 1, 3

[24] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *NeurIPS*, 2020. 3

[25] Han Zhao, Remi Tachet des Combes, Kun Zhang, and Geoffrey J. Gordon. On learning invariant representations for domain adaptation. In *ICML*, 2019. 1, 3, 7